MIL-M-28787/114A(EC)
22 December 1981
SUPERSEDING
MIL-M-28787/114(NAVY)
18 January 1974

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE BDA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

THIS SPECIFICATION INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN. ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.70 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BDA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/114-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

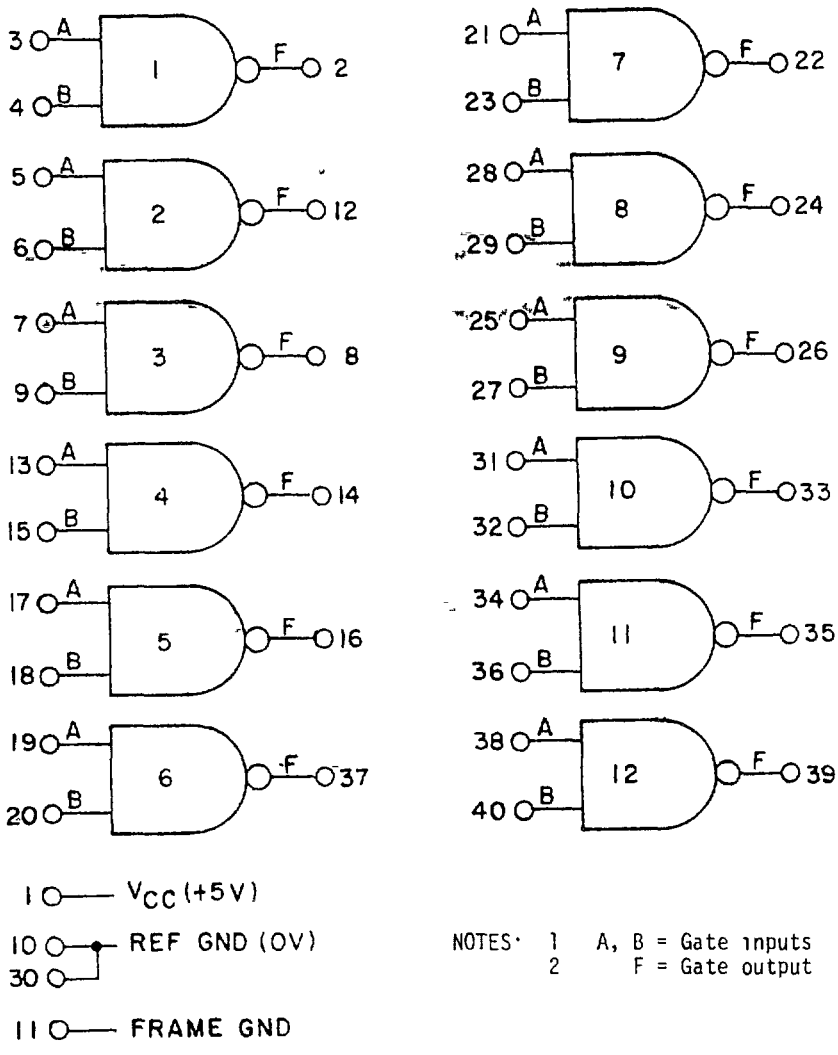
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES: 1 A, B = Gate inputs
2 F = Gate output

FIGURE 1 BDA module function diagram, twelve 2-input NAND gates

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A24	Zener applied potential, cond A	III	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
B1-B12	Zener applied potential, cond B	III	S	0.0	0.40	0.0	0.44	0.0	0.42	0.0	0.46	V
C1-C12	V_{OUT} (low loaded)	IV	S	0.0	1.1	0.0	1.1	0.0	1.1	0.0	1.1	V
D1	Isolation	V	S	10	∞	10	∞	10	∞	10	∞	MΩ
D2	Continuity	V	S	0	1	0	1	0	1	0	1	Ω
E1-E24	Output voltage, high level	VI	X	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
F1-F12	Output voltage, low level	VI	X	0.0	0.40	0.0	0.44	0.0	0.42	0.0	0.46	V
G1-G12	Output short circuit current	VII	X	0.70	1.31	0.70	1.31	0.70	1.31	0.70	1.31	mA
H1-H24	Input current, low level	VIII	X	0.2	1.6	0.2	1.6	0.2	1.6	0.2	1.6	mA
J1-J24	Input current, high level	VIII	X	0	2	0	4	0	3	0	5	µA
K1-K24	Input capacitance	IX	Q	2	16	2	16	2	16	2	16	pF
L1	Power supply current, high level	X	S	0	33	0	33	0	33	0	33	mA
L2	Power supply current, low level	X	S	0	53	0	53	0	53	0	53	mA
M	Power supply transient rise time	XI	S	36	148	36	148	36	148	36	148	µs

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N1-N12	Output pulse amplitude, high level	XII	S	3.8	5.5	3.5	5.5	3.8	5.5	3.5	5.5	V
N13-N24	Output pulse amplitude, low level	XII	S	0.0	0.40	0.0	0.44	0.0	0.42	0.0	0.46	V
P1-P12	Transition time (t_{THL})	XIII	S	6	24	6	24	6	24	6	24	ns
P13-P24	Transition time (t_{TLH})	XIII	S	60	160	60	160	60	160	60	160	ns
Q1-Q12	Propagation delay time (t_{PLH})	XIV	S	20	75	20	75	20	75	20	75	ns
Q13-Q24	Propagation delay time (t_{PHL})	XIV	S	6	40	6	40	6	40	6	40	ns
R1-R12	Output crosstalk, high level	XV	Q	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V
S1-S12	Output crosstalk low level	XV	Q	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be 25°C ambient sample acceptance tested (one percent or two modules from each inspection lot, whichever is greater).

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through S shall be performed in any order with the following exceptions. If test conditions A, B, C or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D-where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through S in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions E through J and L through S shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests E through J and L through S shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests E through S shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military specification MIL-M-28787/114 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2658634C dated March 15, 1971. Part number M28787/114-1 supersedes part numbers 2658634 and M28787/114. These parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	
(all outputs at low level)	292 mW (outputs not loaded)
(all outputs at high level)	182 mW (outputs not loaded)
Supply voltage	5.0V DC ±10 percent
Input voltage	
High level	1.8V minimum, V_{CC} maximum
Low level	-0.5V minimum, 1.2V maximum

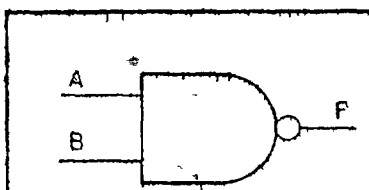
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Input current	
High level	50 µA maximum
Low level	-1.6 mA maximum
Output voltage	
High level	3.5V minimum V_{CC} maximum
Low level	0.0V minimum, 0.46V maximum
Output drive current	
High level	-60 µA maximum
Low level	11.2 mA maximum

6.5 Module function. This module consists of twelve independent 2-input NAND gates.

6.5.1 Input to output relationship. Each NAND gate shall have the input to output relationship for positive logic as shown in table II.

TABLE II. Input to output relationship for each NAND gate



Inputs		Output
A	B	F
L	L	H
L	H	H
H	L	H
H	H	L

1/ A, B = Gate inputs.
2/ F = Gate output.
3/ H = High level voltage.
4/ L = Low level voltage.

6.5.2 Parallel of outputs. The paralleling of outputs is permissible, however, this produces an increase in high level drive capability and a decrease in low level drive capability. This increase and decrease is calculated from the following equations:

Reduced low level drive capability = Maximum drive - (5/6) (n-1)

where maximum drive = 7 S.L.L. (Standard Logic Loads, Low level) and n = the number of outputs connected in parallel.

Increased high level drive capability = (maximum) (n)

where maximum drive = 12 S.L.L. (Standard Logic Loads, High level) and n = the number of outputs connected in parallel.

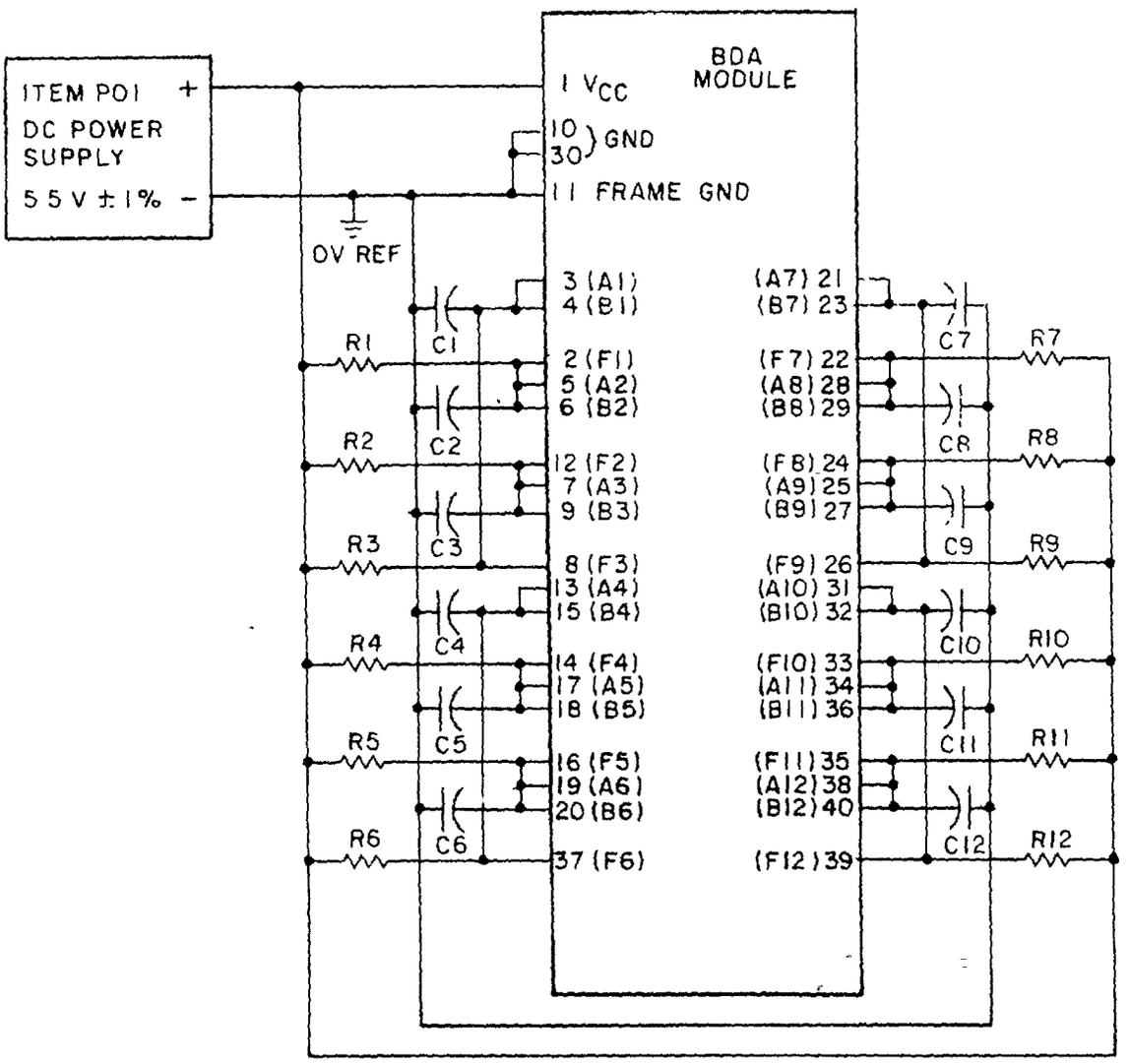
6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes

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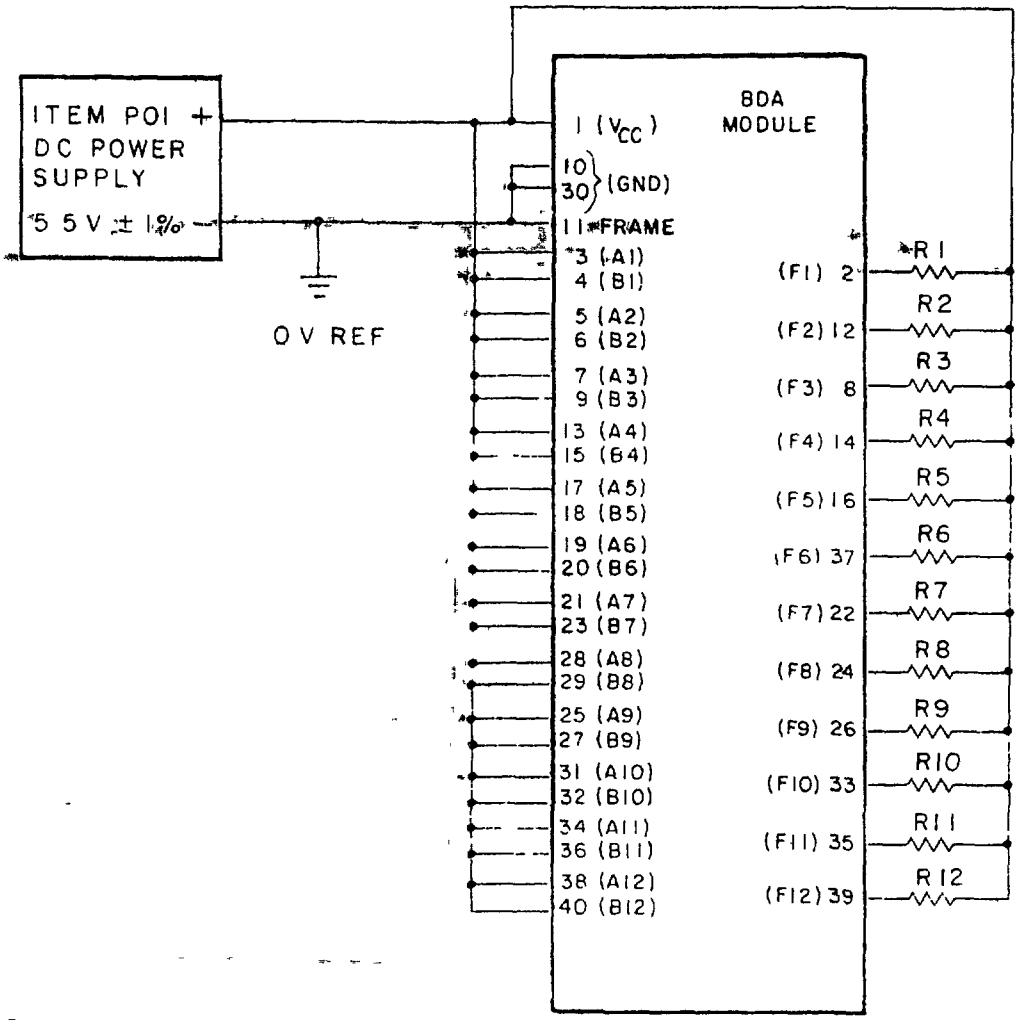
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NOTES: 1. R1-R12 = 422 Ω ±1%, 1/8 W
2. C1-C12 = 220 pF ±20%.
3. Capacitors not required for life test

FIGURE 2 Operating temperature and life test circuit.

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NOTE R1-R12 = 422 Ω ±1%, 1/8 W.

FIGURE 3 Component temperature test circuit

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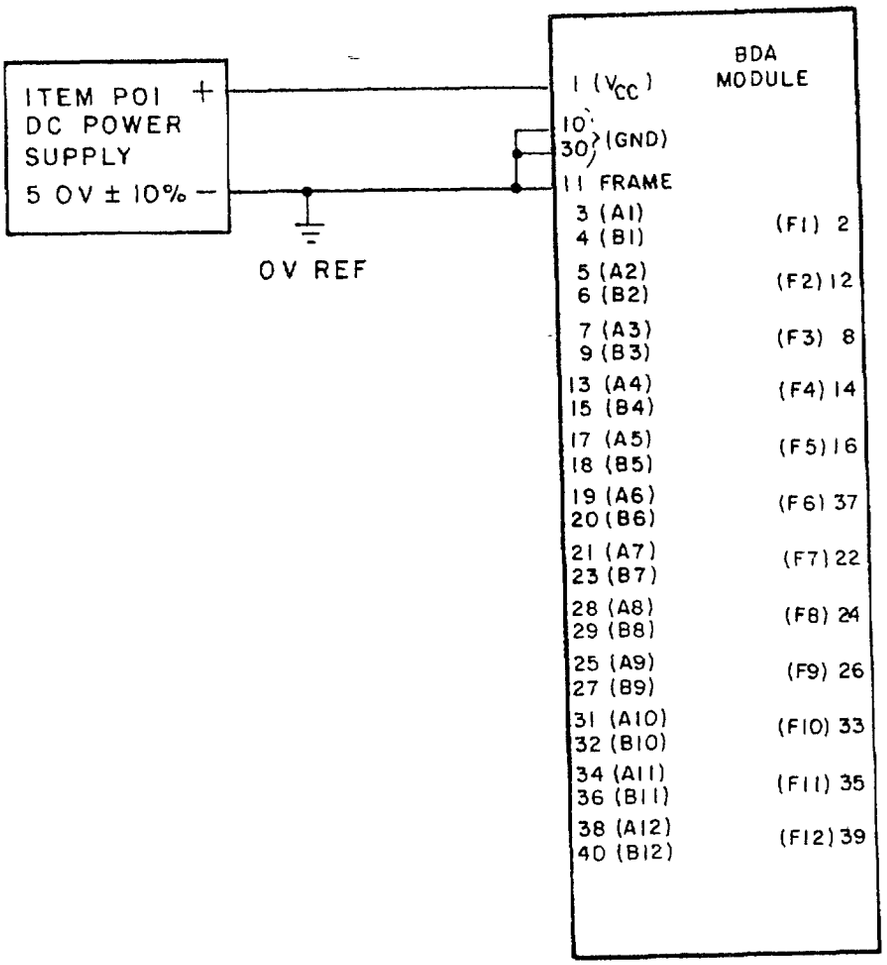


FIGURE 4 Vibration test circuit.

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TABLE III. Zener applied potential test.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
A1	F1 Zener applied potential, cond A	Open	2	4	1	3	3.8	5.5	V	5
A2	F1		2	3		4				
A3	F2		12	6		5				
A4	F2		12	5		6				
A5	F3		8	9		7				
A6	F3		8	7		9				
A7	F4		14	15		13				
A8	F4		14	13		15				
A9	F5		16	18		17				
A10	F5		16	17		18				
A11	F6		37	20		19				
A12	F6		37	19		20				
A13	F7		22	23		21				
A14	F7		22	21		23				
A15	F8		24	29		28				
A16	F8		24	28		29				
A17	F9		26	27		25				

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TABLE III. Zener applied potential test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
A18	F9 Zener applied potential, cond A	Open	26	25	1	27	3.8	5.5	V	5
A19	F10		33	32		31				
A20	F10		33	31		32				
A21	F11		35	36		34				
A22	F11		35	34		36				
A23	F12		39	40		38				
A24	F12		39	38		40				
B1	F1 Zener applied potential, cond B	1	2	Open	Open	3	0.0	0.40		
B2	F2		12			5				
B3	F3		8			7				
B4	F4		14			13				
B5	F5		16			17				
B6	F6		37							
B7	F7		22			21				
B8	F8		24			28				
B9	F9		26			25				
B10	F10		33			31				
B11	F11		35			34				
B12	F12		39			38				

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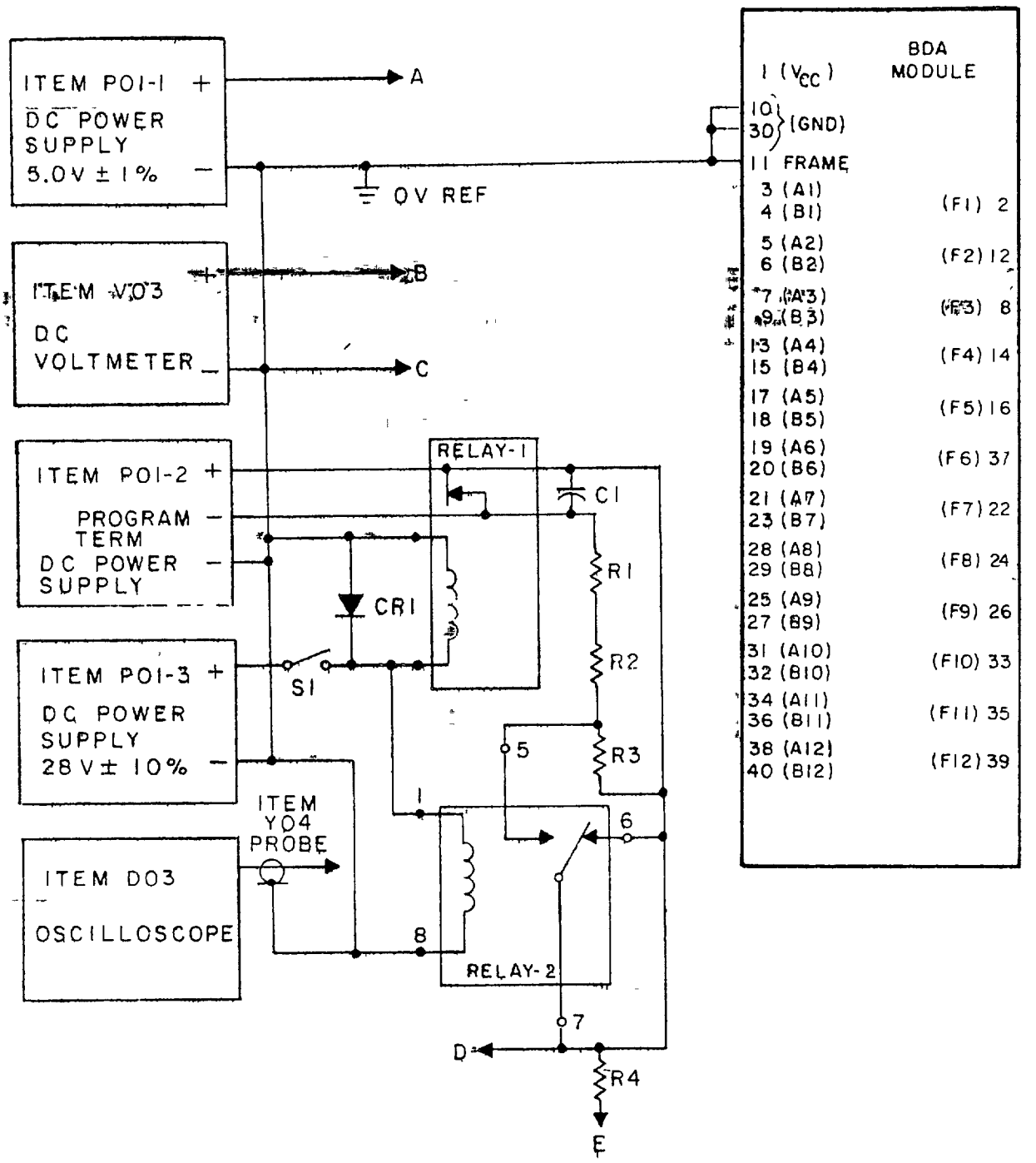
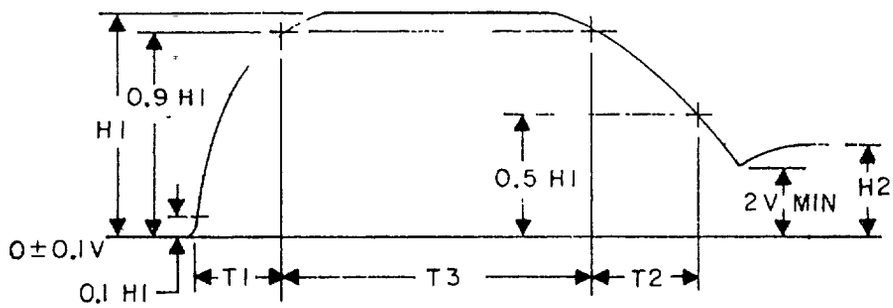


FIGURE 5. Zener applied potential test circuit for table III.

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NOTES:
1. R1 = 178 Ω ±1%, 1/8 W
2. R2 = 301 Ω ±1%, 1/8 W
3. R3 = 649 Ω ±1%, 1/8 W
4. R4 = 154 Ω ±1%, 1/2 W
5. C1 = 22 µF ±10%
6. CR1 = 1N647 or equivalent
7. Relay-1 = 28 volt
8. Relay-2 = Agastat 2112 or equivalent
9. Resistors R1, R2, R3, and capacitor C1 are programming components for power supply PO1-2 and are based on a programming constant of 100 ohms per volt.
10. Using item D03 with item Y04, adjust the time delay relay to obtain the following output waveform at pin 7 of the time delay relay after switch S1 is closed
11. For each test condition, switch S1 is closed and the voltage reading is made a minimum of 2 seconds after the closure of switch S1. At the conclusion of each test condition measurement, switch S1 is returned to the open position.
12. This circuit is drawn showing power supply PO1-2 with a negative reference terminal (non-commoned programming terminal labeled (-)). This circuit, as shown, will not operate properly if power supply PO1-2 has a positive non-commoned programming terminal.
NOTE: Load current must not pass through the sensing network.



Output waveform parameters

H1 = 11.25V ±1%.
H2 = 5.0V ±10%.
T1 = 100 ms +100, -50 ms.
T2 = 100 ms +400, -50 ms.
T3 = 0.5 sec ±10%.

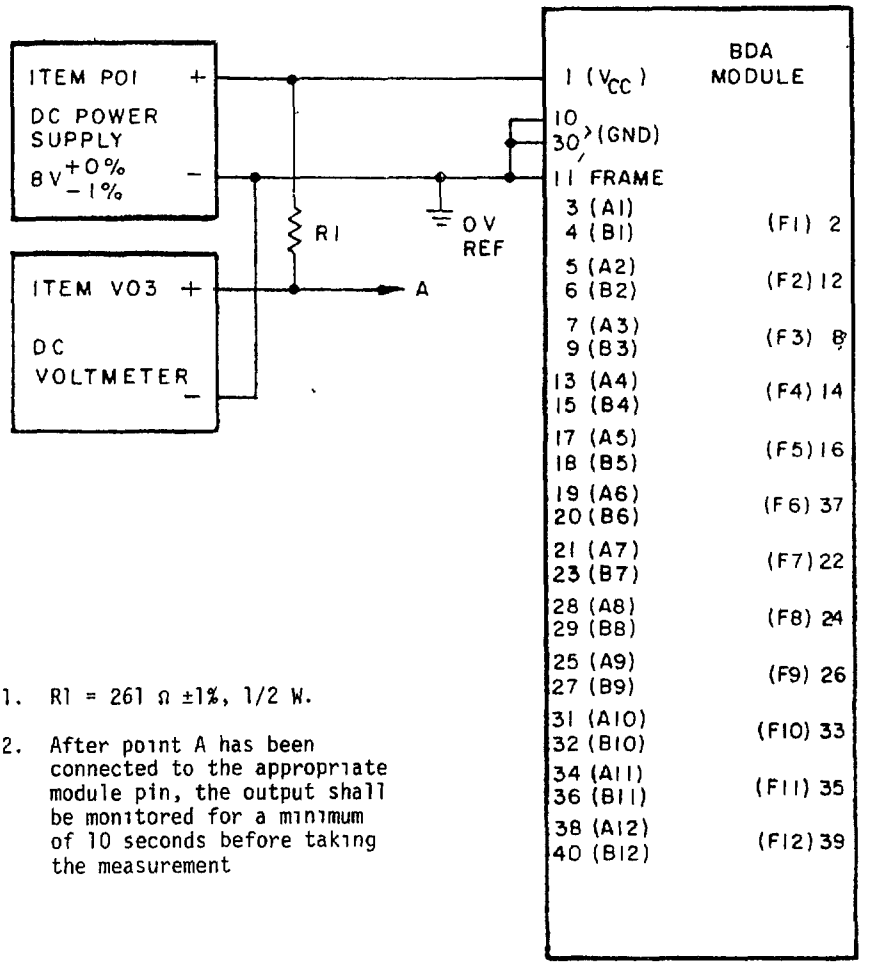
FIGURE 5 Zener applied potential test circuit for table III - Continued

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TABLE IV. V_{OUT} (low loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	F1 V_{OUT} (low loaded)	2	0.0	1.1	V	6
C2	F2	12	↓	↓	↓	↓
C3	F3	8				
C4	F4	14				
C5	F5	16				
C6	F6	37				
C7	F7	22				
C8	F8	24				
C9	F9	26				
C10	F10	33				
C11	F11	35				
C12	F12	39				

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NOTES 1. R1 = 261 Ω ±1%, 1/2 W.

2. After point A has been connected to the appropriate module pin, the output shall be monitored for a minimum of 10 seconds before taking the measurement

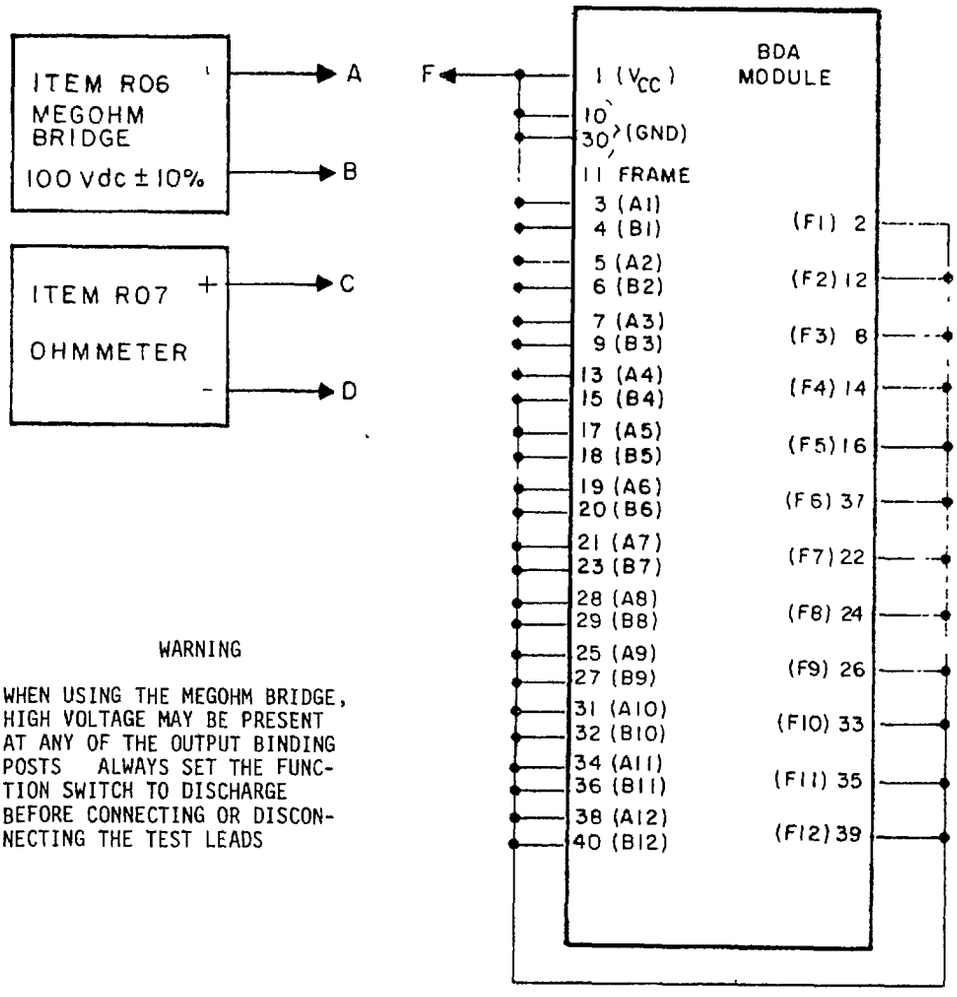
FIGURE 6 V_{OUT} (low loaded) test circuit for table IV

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TABLE V. Isolation and continuity test

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	Isolation	11	F	Open	Open	10	∞	MΩ	7
D2	Continuity	Open	Open	11	Frame	0	1	Ω	7

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WARNING

WHEN USING THE MEGOHM BRIDGE, HIGH VOLTAGE MAY BE PRESENT AT ANY OF THE OUTPUT BINDING POSTS ALWAYS SET THE FUNCTION SWITCH TO DISCHARGE BEFORE CONNECTING OR DISCONNECTING THE TEST LEADS

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead

FIGURE 7 Isolation and continuity test circuit for table V.

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TABLE VI. Output voltage (high-level and low-level) test.

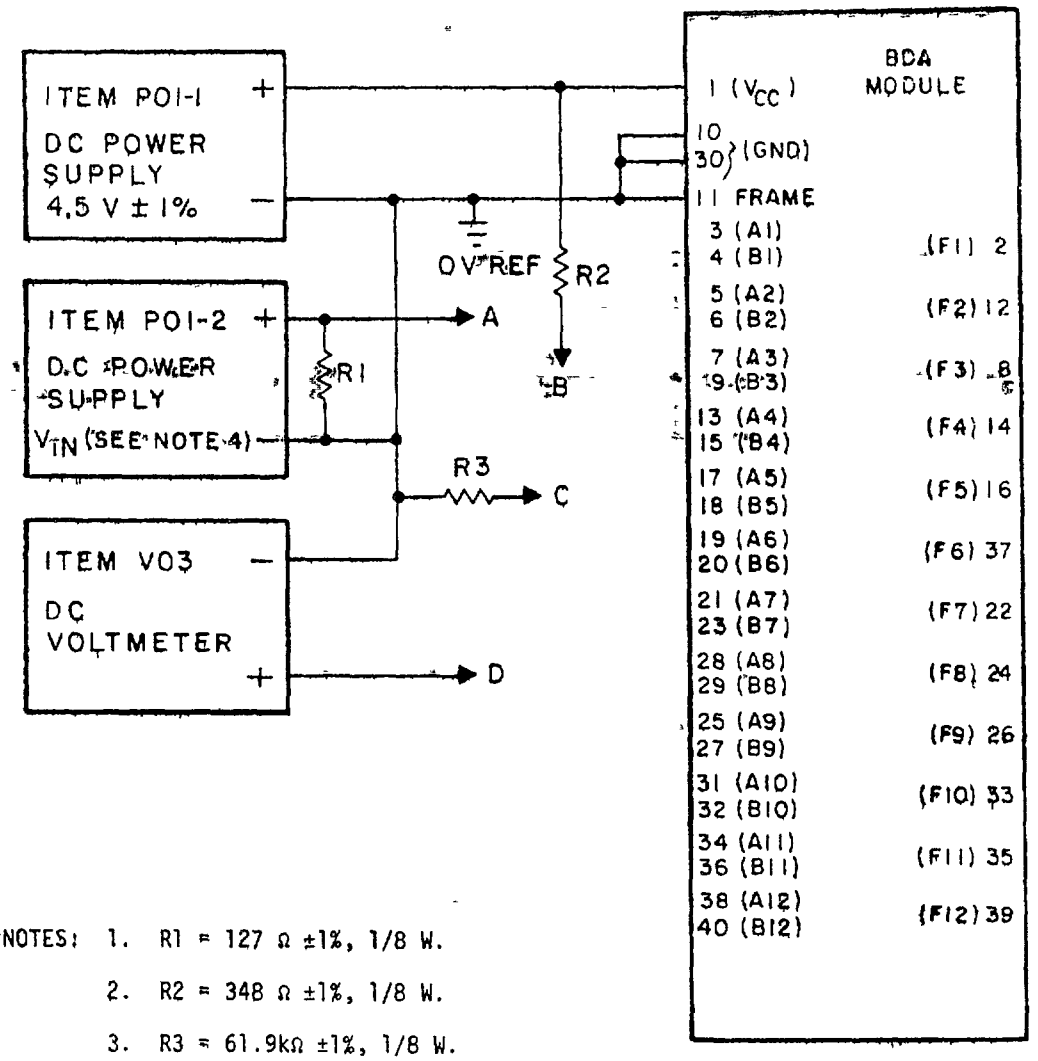
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	F1 Output voltage, high level	3	Open	2	2	3.8	5.5	V	8
E2	F1	4		2	2				
E3	F2	5		12	12				
E4	F2	6		12	12				
E5	F3	7		8	8				
E6	F3	9		8	8				
E7	F4	13		14	14				
E8	F4	15		14	14				
E9	F5	17		16	16				
E10	F5	18		16	16				
E11	F6	19		37	37				
E12	F6	20		37	37				
E13	F7	21		22	22				
E14	F7	23		22	22				
E15	F8	28		24	24				
E16	F8	29		24	24				
E17	F9	25		26	26				
E18	F9	27		26	26				
E19	F10	31		33	33				
E20	F10	32		33	33				
E21	F11	34		35	35				
E22	F11	36		35	35				
E23	F12	38		39	39				
E24	F12	40		39	39				

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TABLE VI. Output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	F1 Output voltage, low level	3, 4	2	Open	2	0.0	0.40	V	8
F2	F2	5, 6	12		12				
F3	F3	7, 9	8		8				
F4	F4	13,15	14		14				
F5	F5	17,18	16		16				
F6	F6	19,20	37		37				
F7	F7	21,23	22		22				
F8	F8	28,29	24		24				
F9	F9	25,27	26		26				
F10	F10	31,32	33		33				
F11	F11	34,36	35		35				
F12	F12	38,40	39		39				

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NOTES: 1. R1 = 127 Ω ±1%, 1/8 W.

2. R2 = 348 Ω ±1%, 1/8 W.

3. R3 = 61.9kΩ ±1%, 1/8 W.

4. Use appropriate V_{IN} for specified test condition and temperature listed below:

Test condition	Temperature (°C)	V_{IN} (volts)
E1-E24	0	1.20
	25	1.10
	60	0.95
	80	0.80
F1-F12	0	2.0
	25	1.9
	60	1.8
	80	1.8

0°, 60°, and 80°C refer to operating temperatures (module fin); 25°C refers to ambient temperature. For 80°C use the 0° to 60°C end-of-life limits.

FIGURE 8 Output voltage (high-level and low-level) test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	F1 Output short circuit current	Open	2	3, 4	0.70	1.31	mA	9
G2	F2		12	5, 6				
G3	F3		8	7, 9				
G4	F4		14	13,15				
G5	F5		16	17,18				
G6	F6		37	19,20				
G7	F7		22	21,23				
G8	F8		24	28,29				
G9	F9		26	25,27				
G10	F10		33	31,32				
G11	F11		35	34,36				
G12	F12		39	38,40				

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TABLE VIII. Input current (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	A1 Input current, low level	Open	3	Open	0.2	1.6	mA	9
H2	B1		4					
H3	A2		5					
H4	B2		6					
H5	A3		7					
H6	B3		9					
H7	A4		13					
H8	B4		15					
H9	A5		17					
H10	B5		18					
H11	A6		19					
H12	B6		20					
H13	A7		21					
H14	B7		23					
H15	A8		28					
H16	B8		29					
H17	A9		25					

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TABLE VIII. Input current (low-level and high-level) test - Continued.

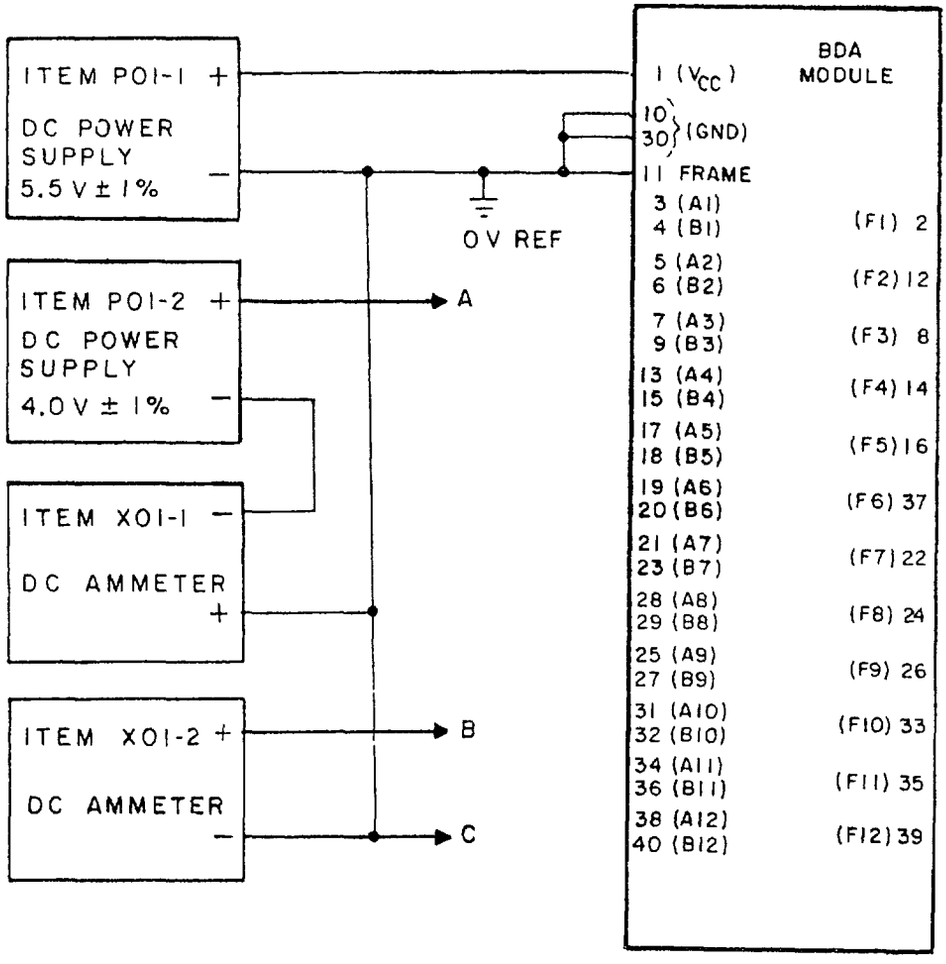
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
H18	B9 Input current, low level	Open	27	Open	0.2	1.6	mA	9
H19	A10		31					
H20	B10		32					
H21	A11		34					
H22	B11		36					
H23	A12		38					
H24	B12		40					
J1	A1 Input current, high level	3	Open	4	0	2	µA	
J2	B1	4		3				
J3	A2	5		6				
J4	B2	6		5				
J5	A3	7		9				
J6	B3	9		7				
J7	A4	13		15				
J8	B4	15		13				
J9	A5	17		18				

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TABLE VIII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J10	B5 Input current, high level	18	Open	17	0	2	µA	9
J11	A6	19		20				
J12	B6	20		19				
J13	A7	21		23				
J14	B7	23		21				
J15	A8	28		29				
J16	B8	29		28				
J17	A9	25		27				
J18	B9	27		25				
J19	A10	31		32				
J20	B10	32		31				
J21	A11	34		36				
J22	B11	36		34				
J23	A12	38		40				
J24	B12	40		38				

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CAUTION

DO NOT ISOLATE CHASSIS OF XO1 AMMETER FROM EARTH GROUND

CAUTION

THE CHASSIS OF POWER SUPPLY PO1-2 SHALL BE ISOLATED FROM EARTH GROUND

NOTE: Since ammeters XO1-1 and XO1-2 are not used simultaneously, only one is required.

FIGURE 9 Output short circuit current and input current (low-level and high-level) test circuit for tables VII and VIII.

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TABLE IX. Input capacitance test.

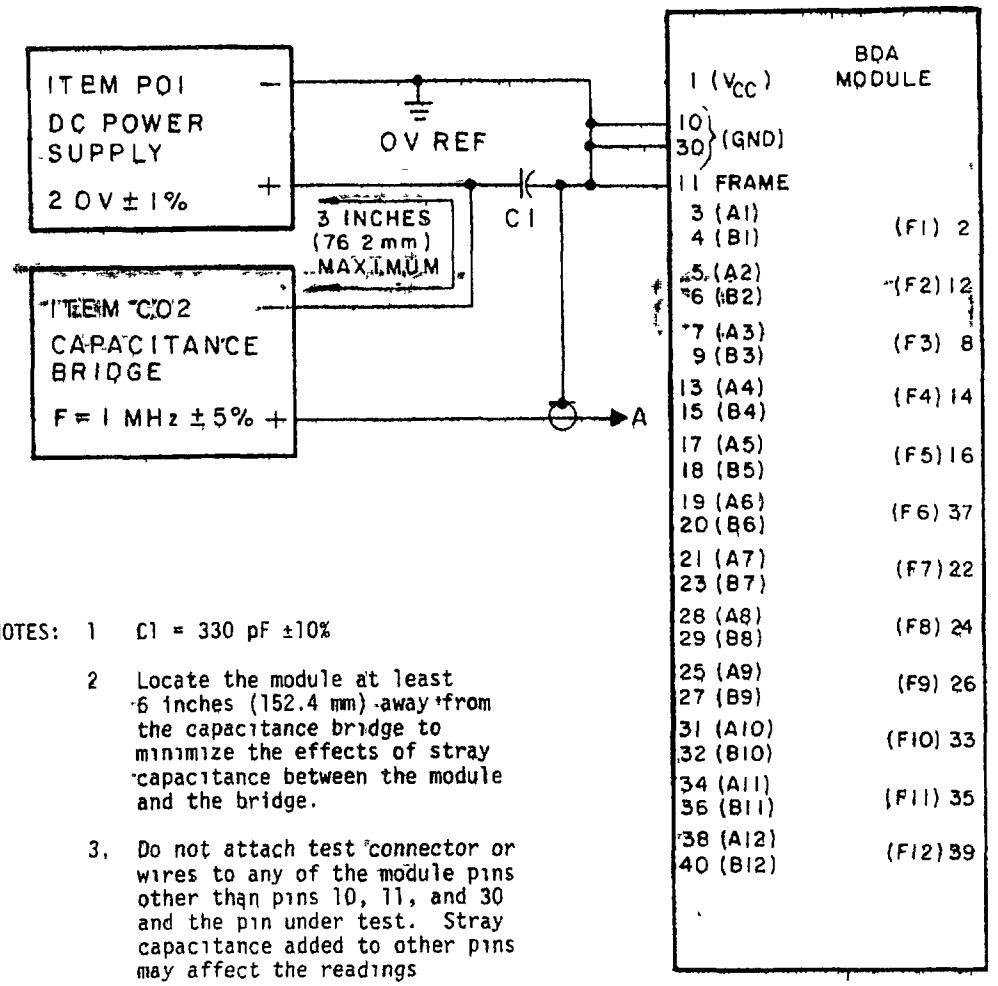
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
K1	A1 Input capacitance	3	2	16	pF	10
K2	B1	4				
K3	A2	5				
K4	B2	6				
K5	A3	7				
K6	B3	9				
K7	A4	13				
K8	B4	15				
K9	A5	17				
K10	B5	18				
K11	A6	19				
K12	B6	20				
K13	A7	21				

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TABLE IX. Input capacitance test - Continued.

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
K14	B7 Input capacitance	23	2	16	pF	10
K15	A8	28				
K16	B8	29				
K17	A9	25				
K18	B9	27				
K19	A10	31				
K20	B10	32				
K21	A11	34				
K22	B11	36				
K23	A12	38				
K24	B12	40				

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NOTES: 1. C1 = 330 pF ±10%

2. Locate the module at least 6 inches (152.4 mm) away from the capacitance bridge to minimize the effects of stray capacitance between the module and the bridge.

3. Do not attach test connector or wires to any of the module pins other than pins 10, 11, and 30 and the pin under test. Stray capacitance added to other pins may affect the readings

4. Null out the effects of the test leads prior to connecting to the module

5. The output voltage of the capacitance bridge CO2 shall not exceed 0.5 Vrms.

FIGURE 10 Input capacitance test circuit for table IX.

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TABLE X. Power supply current (high-level and low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L1	Power supply current, high level	10	0	33	mA	11
L2	Power supply current, low level	1	0	53	mA	11

TABLE XI. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M	Power supply transient rise time	36	148	µs	12, 13

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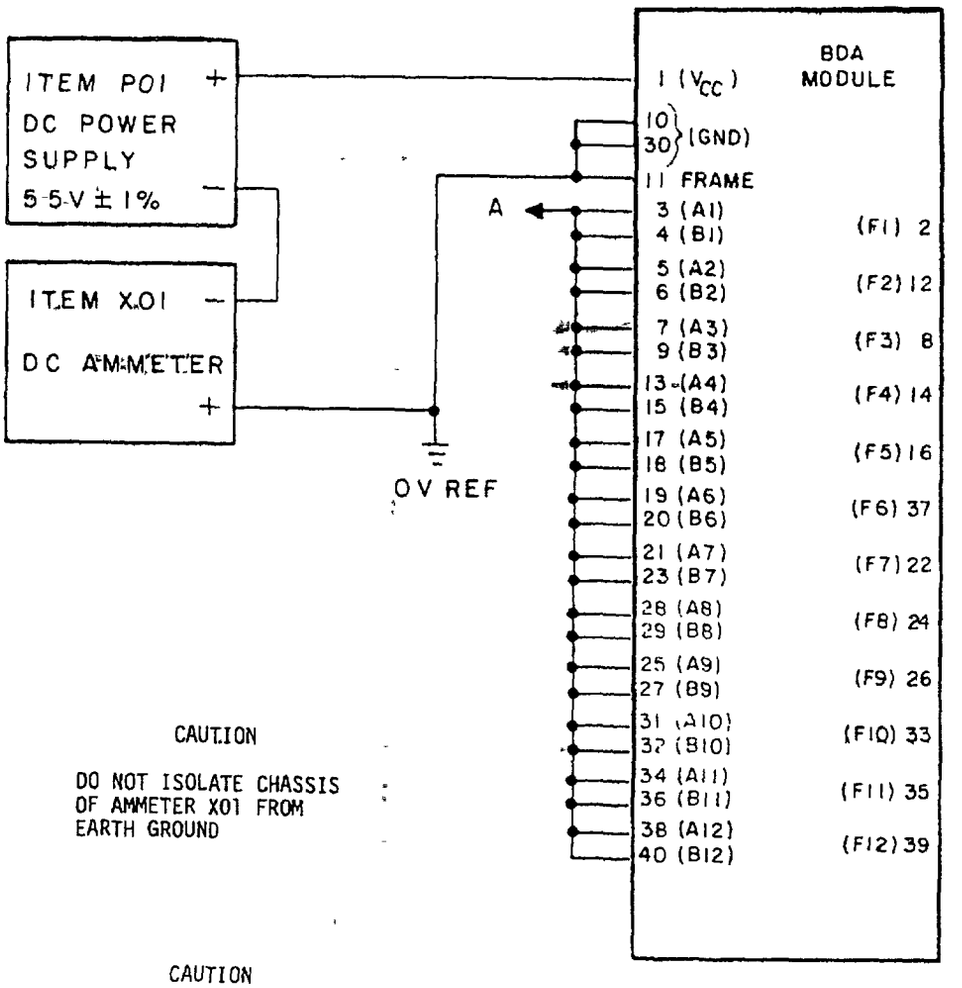
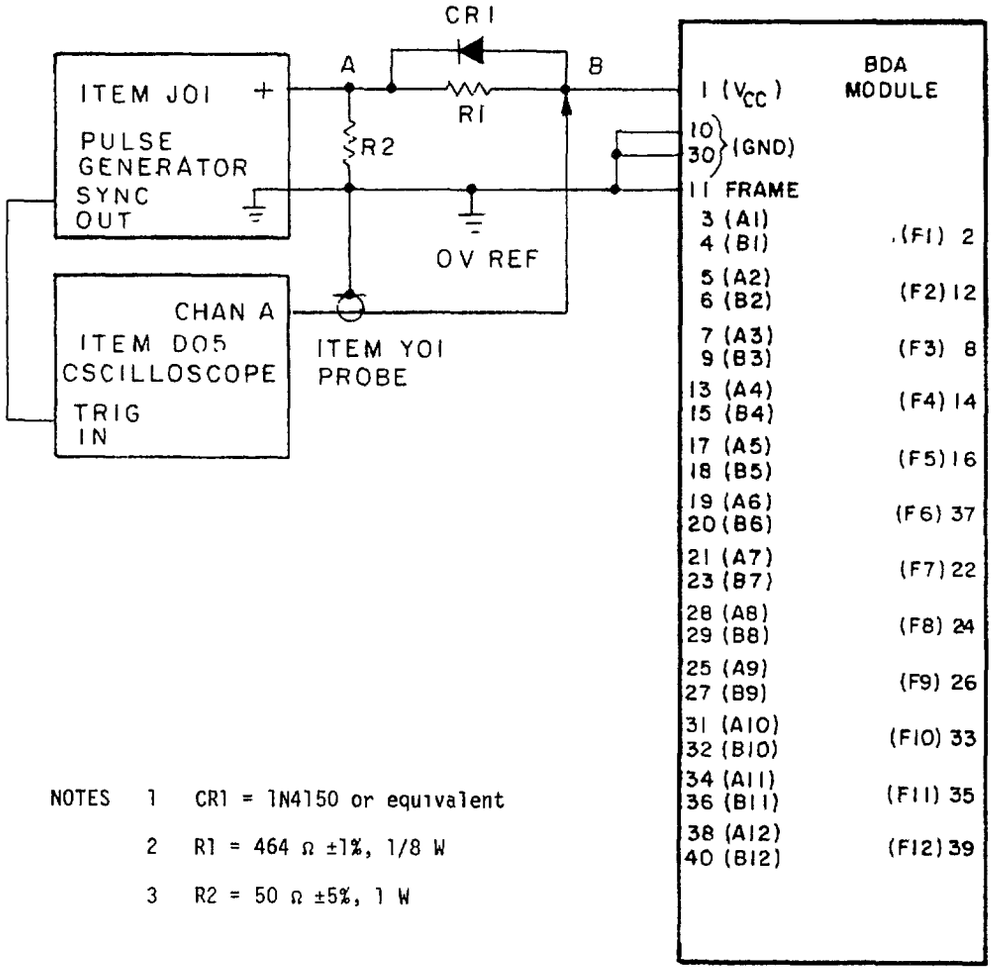


FIGURE 11 Power supply current (high-level and low-level) test circuit for table X.

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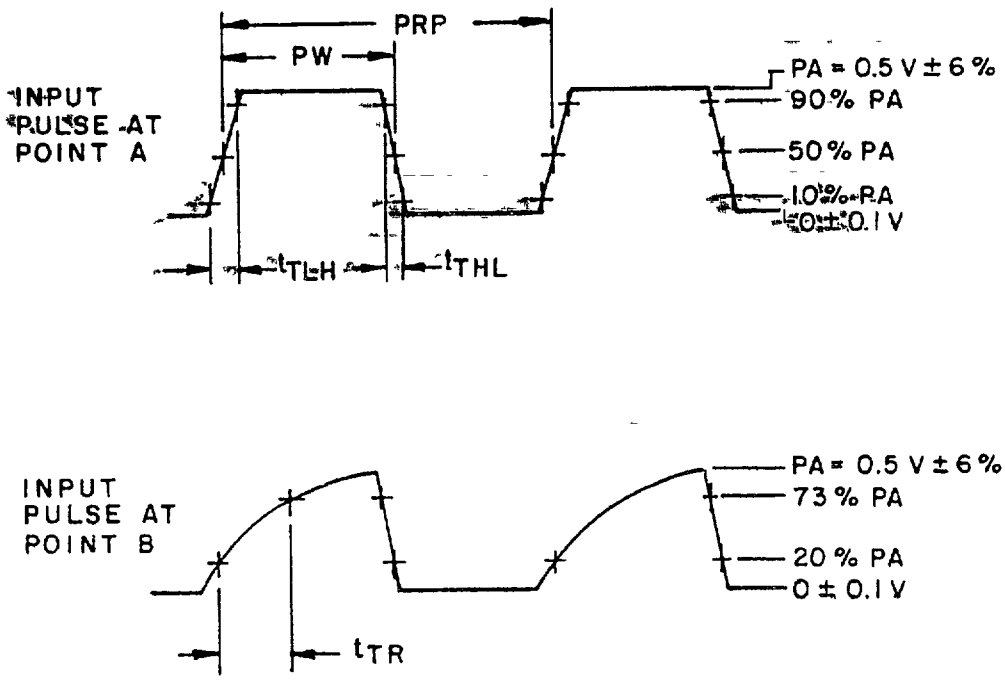
NOTES 1 CR1 = 1N4150 or equivalent

2 R1 = 464 Ω ±1%, 1/8 W

3 R2 = 50 Ω ±5%, 1 W

FIGURE 12 Power supply transient rise time test circuit for table XI

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NOTE Input pulse setup:

PW = 750 µs ±5%.

PRP = 1 5 ms ±10%.

$t_{THL} = t_{TLH}$ = 50 ns ±10%.

FIGURE 13. Power supply transient rise time test circuit waveforms for table XI.

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TABLE XII. Output pulse amplitude (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	F1 Output pulse amplitude, high level	3	4	2	3.8	5.5	V	14,15
N2	F2	5	6	12				
N3	F3	7	9	8				
N4	F4	15	13	14				
N5	F5	18	17	16				
N6	F6	20	19	37				
N7	F7	21	23	22				
N8	F8	28	29	24				
N9	F9	25	27	26				
N10	F10	31	32	33				
N11	F11	34	36	35				
N12	F12	40	38	39				
N13	F1 Output pulse amplitude, low level	3	4	2	0.0	0.40		

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TABLE XII. Output pulse amplitude (high-level and low-level) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
N14	F2 Output pulse amplitude, low level	5	6	12	0.0	0.40	V	14, 15
N15	F3	7	9	8				
N16	F4	15	13	14				
N17	F5	18	17	16				
N18	F6	20	19	37				
N19	F7	21	23	22				
N20	F8	28	29	24				
N21	F9	25	27	26				
N22	F10	31	32	33				
N23	F11	34	36	35				
N24	F12	40	38	39				

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TABLE XIII. Transition time (t_{THL} and t_{TLH}) test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect D to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
P1	F1 Transition time, (t_{THL})	3	4	2	6	24	ns	14,15
P2	F2	5	6	12				
P3	F3	7	9	8				
P4	F4	15	13	14				
P5	F5	18	17	16				
P6	F6	20	19	37				
P7	F7	21	23	22				
P8	F8	28	29	24				
P9	F9	25	27	26				
P10	F10	31	32	33				
P11	F11	34	36	35				
P12	F12	40	38	39				
P13	F1 Transition time, (t_{TLH})	3	4	2	60	160		

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TABLE XIII. Transition time (t_{THL} and t_{TLH}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	D to pin	Min	Max		
P14	F2 Transition time, (t_{TLH})	5	6	12	60	160	ns	14;15
P15	F3	7	9	8				
P16	F4	15	13	14				
P17	F5	18	17	16				
P18	F6	20	19	37				
P19	F7	21	23	22				
P20	F8	28	29	24				
P21	F9	25	27	26				
P22	F10	31	32	33				
P23	F11	34	36	35				
P24	F12	40	38	39				

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test.

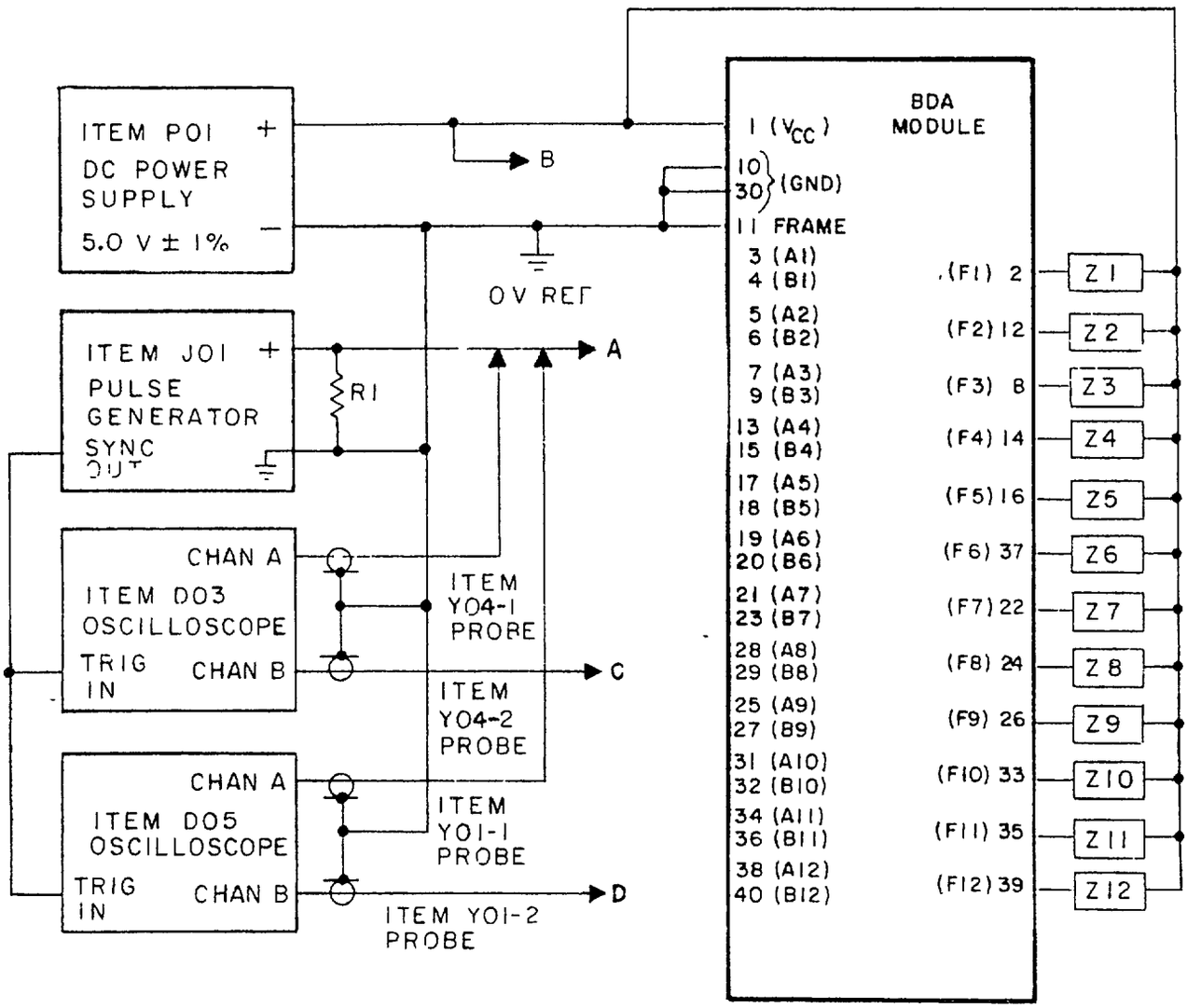
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	D to pin	Min	Max		
Q1	F1 Propagation delay time (t_{PLH})	3	4	2	20	75	ns	14,15
Q2	F2	5	6	12				
Q3	F3	7	9	8				
Q4	F4	15	13	14				
Q5	F5	18	17	16				
Q6	F6	20	19	37				
Q7	F7	21	23	22				
Q8	F8	28	29	24				
Q9	F9	25	27	26				
Q10	F10	31	32	33				
Q11	F11	34	36	35				
Q12	F12	40	38	39				
Q13	F1 Propagation delay time (t_{PHL})	3	4	2	6	40		
Q14	F2	5	6	12				
Q15	F3	7	9	8				
Q16	F4	15	13	14				
Q17	F5	18	17	16				

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TABLE XIV. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q18	F6 Propagation delay time (t_{PHL})	20	19	37	6	40	ns	14,15
Q19	F7	21	23	22				
Q20	F8	28	29	24				
Q21	F9	25	27	26				
Q22	F10	31	32	33				
Q23	F11	34	36	35				
Q24	F12	40	38	39				

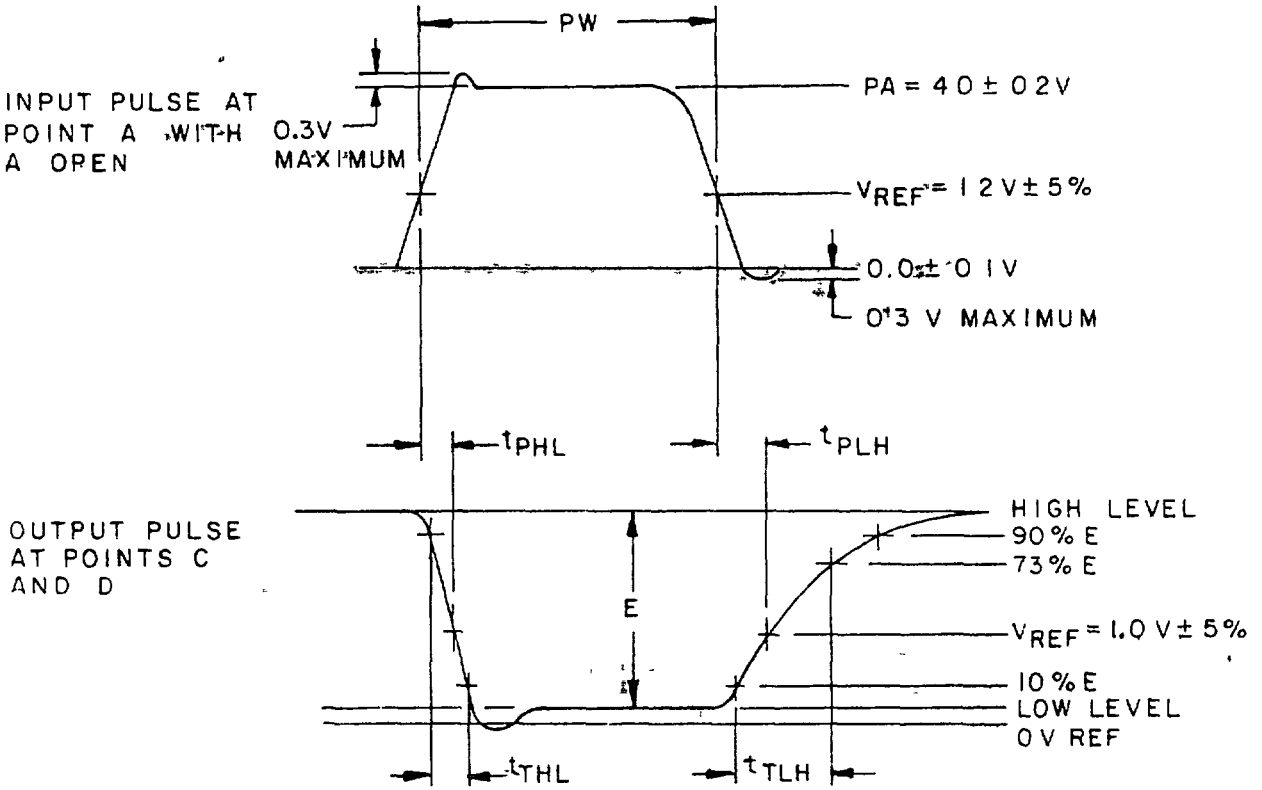
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NOTES: 1. R1 = 50 Ω ±5%, 1 W
2. Z1-Z12 = use load circuit 1 per figure 16.

FIGURE 14. Output pulse amplitude (high-level and low-level), transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XII, XIII and XIV.

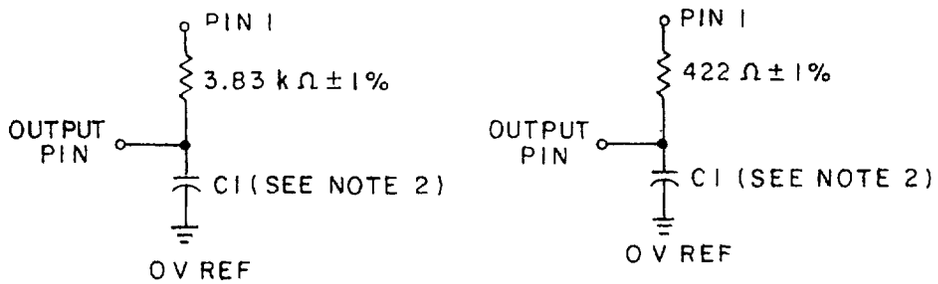
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NOTE Input pulse parameters
PW = 1 µs ±10%
$t_{TLH} = t_{THL}$ = 20 ±2 ns
PRR = 100 kHz ±10%

FIGURE 15. Output pulse amplitude (high-level and low-level), transition time (t_{THL} and t_{TLH}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII, XIII and XIV

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LOAD CIRCUIT 1 LOAD CIRCUIT 2

NOTES 1 The loop length from pin 1 to each load resistor and output, including the length of the body of the resistor shall not exceed 4 5 inches (114 3 mm)

2 With the module removed from the test connector, using capacitance bridge C02, measure the capacitance between the test connector pins listed in the following tabulation The measurements shall be performed at the point of mechanical contact between the module blade pin and the test connector pin The measured values at 1 MHz ±10% shall be as specified herein. These measurements are made with the oscilloscope probe disconnected and the 3.83 kΩ resistors disconnected from their respective output pins The resistors should be disconnected as close to their bodies as practical. Reconnect these resistors prior to using the test connector for module testing

Test connector pins		Total capacitance 1/	
From pin	To pin	Load circuit 1	Load circuit 2
2 12 8 14 16 37 22 24 26 33 35 39	10,11,30	C1 = 30 ±1 pF	C1 = 30 ±5 pF

1/ Total capacitance includes measured capacitance and probe capacitance If necessary, install a compensating capacitor directly at the test connector pins to obtain this value.

FIGURE 16. BDA logic load circuit

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TABLE XV. Output crosstalk (high-level and low-level) test.

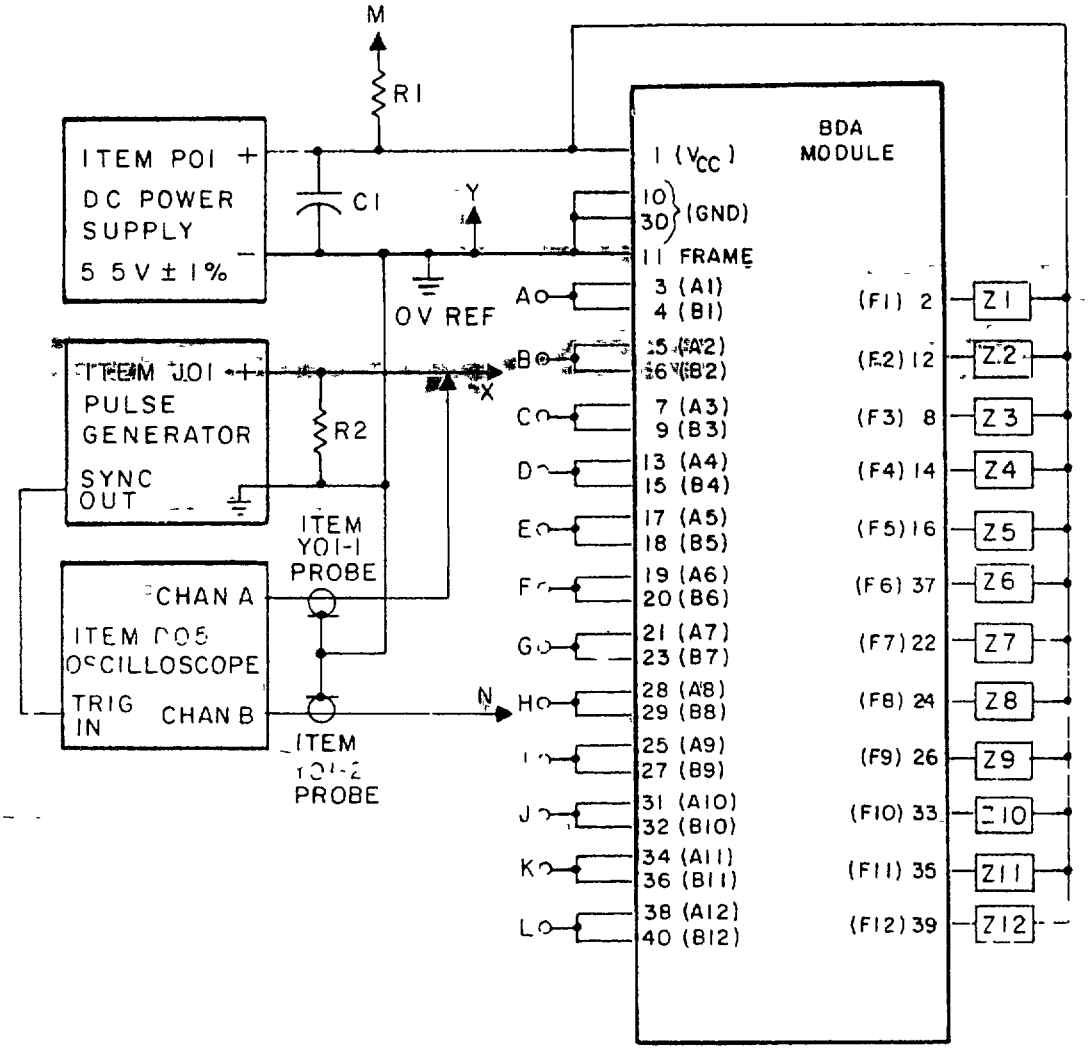
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		X to points	Y to point	M to point	N to point	Min	Max		
R1	F1 Output crosstalk, high level	B,C,D,E,F, G,H,I,J,K,L	A	Open	2	0.0	0.8	V	17,18
R2	F2	A,C,D,E,F, G,H,I,J,K,L	B		12				
R3	F3	A,B,D,E,F, G,H,I,J,K,L	C		8				
R4	F4	A,B,C,E,F, G,H,I,J,K,L	D		14				
R5	F5	A,B,C,D,F, G,H,I,J,K,L	E		16				
R6	F6	A,B,C,D,E, G,H,I,J,K,L	F		37				
R7	F7	A,B,C,D,E, F,H,I,J,K,L	G		22				
R8	F8	A,B,C,D,E, F,G,I,J,K,L	H		24				
R9	F9	A,B,C,D,E, F,G,H,J,K,L	I		26				
R10	F10	A,B,C,D,E, F,G,H,I,K,L	J		33				
R11	F11	A,B,C,D,E, F,G,H,I,J,L	K		35				
R12	F12	A,B,C,D,E, F,G,H,I,J,K	L		39				
S1	F1 Output crosstalk, low level	B,C,D,E,F, G,H,I,J,K,L	Open	A	2				

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TABLE XV. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		X to points	Y to point	M to point	N to point	Min	Max		
S2	F2 Output crosstalk, low level	A,C,D,E,F, G,H,I,J,K,L	Open	B	12	0.0	0.8	V	17,18
S3	F3	A,B,D,E,F, G,H,I,J,K,L		C	8				
S4	F4	A,B,C,E,F, G,H,I,J,K,L		D	14				
S5	F5	A,B,C,D,F, G,H,I,J,K,L		E	16				
S6	F6	A,B,C,D,E, G,H,I,J,K,L		F	37				
S7	F7	A,B,C,D,E, F,H,I,J,K,L		G	22				
S8	F8	A,B,C,D,E, F,G,I,J,K,L		H	24				
S9	F9	A,B,C,D,E, F,G,H,J,K,L		I	26				
S10	F10	A,B,C,D,E, F,G,H,I,K,L		J	33				
S11	F11	A,B,C,D,E, F,G,H,I,J,L		K	35				
S12	F12	A,B,C,D,E, F,G,H,I,J,K		L	39				

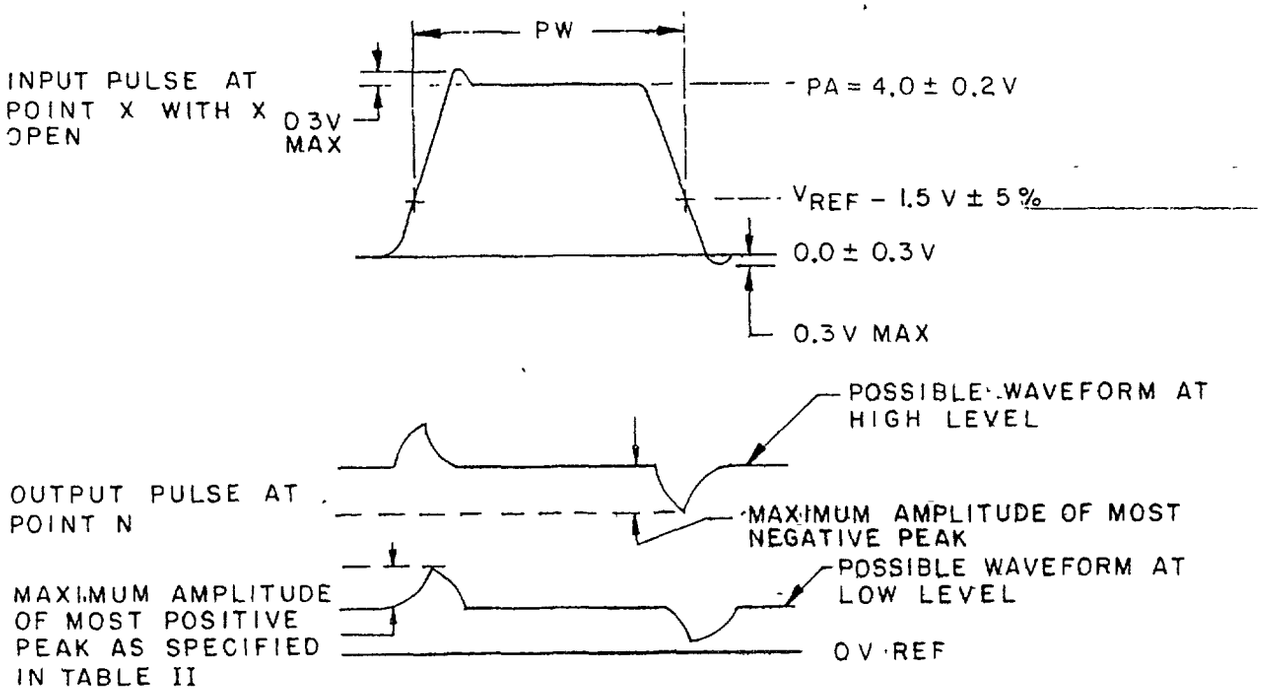
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NOTES
1. R1 = 4 64 kΩ ±1%, 1/8 watt
2. R2 = 50 Ω ±5%, 1 W
3. C1 represents item P01 bypass capacitance added to the test fixture to limit the transient response of the power supply and the power supply transmission paths to module current demands. Capacitance may be added as required
4. Z1-Z12 = use load circuit 2 per figure 16

FIGURE 17 Output crosstalk (high-level and low-level) test circuit for table XV

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NOTE Input pulse parameter
PW = 0.9 ±0.1 µs
$t_{TLH} = t_{THL}$ = 20 ±2 ns.
PRR = 100 kHz ±10%.

FIGURE 18 Output crosstalk (high-level and low-level) test circuit waveforms for table XV.

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